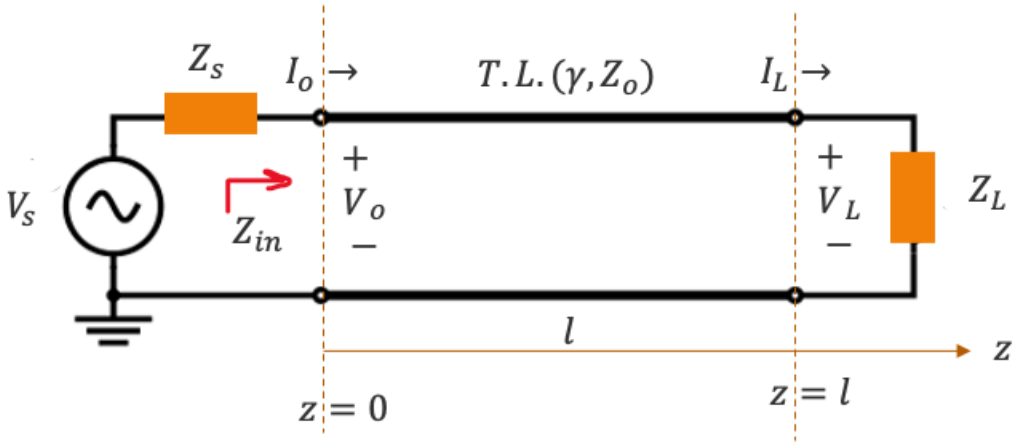
Transmission Lines

The following subjects will be covered

- Wave propagation (guiding) on a TL.
- Analysis and examples.

Analysis of Transmission Line Circuits

Consider the following general circuit with the following axis set-up



To evaluate the voltage reflection $\Gamma(z) = \dfrac{V_o^- \, e^{\gamma z}}{V_o^+ \, e^{-\gamma z}}$ $\Gamma(0) = \dfrac{V_o^-}{V_o^+}$

At load $(z = l)$ $\Gamma_L = \dfrac{Z_L - Z_o}{Z_L + Z_o}$ For a "matched" load $(Z_L = Z_o)$ $\Gamma_L = 0$

At any location $\Gamma(z) = \Gamma_L \, e^{-2\gamma z}$

In general, loads are complex $\Gamma = |\Gamma| \, e^{j\theta_\Gamma}$

Example

Find the reflection coefficient at the load on a 100-Ohm TL at 100 MHz. The load is a 50-Ohm resistor and a 10 pF capacitor in series.

$$Z_L = R_L + 1/j\omega C_L \;\; = 50 - j/(2\pi\times 100\,M)(10\,p)$$

$$\Gamma = \frac{50 - j/(2\pi\times 100\,M)(10\,p) - 100}{50 - \dfrac{j}{(2\pi\times 100\,M)(10\,p)} + 100} = -0.76\, e^{j(119.3°)} \quad = 0.76\angle -60.7°$$

What happens if the resistor is removed (purely reactive load)?

$$\Gamma = \frac{-j/(2\pi\times 100\,M)(10\,p) - 100}{-\dfrac{j}{(2\pi\times 100\,M)(10\,p)} + 100} = -\,e^{j(115.6°)}$$

For any purely reactive load $\quad |\Gamma| = 1 \quad \theta_\Gamma = \tan^{-1}(X_L/Z_o)$

Reflection Analysis

Using the following relation, $V_o^- = \Gamma(0)\, V_o^+$

We write for a lossless line, $V(z) = V_o^+ \left(e^{-j\beta z} + \Gamma\, e^{j\beta z}\right)$ $I(z) = \frac{V_o^+}{Z_o}\left(e^{-j\beta z} - \Gamma\, e^{j\beta z}\right)$

$$|V(z)| = \sqrt{\left[V_o^+ \left(e^{-j\beta z} + \Gamma\, e^{j\beta z}\right)\right] \cdot \left[V_o^+ \left(e^{-j\beta z} + \Gamma\, e^{j\beta z}\right)\right]^*}$$

$$|V(z)| = |V_o^+|\sqrt{1 + |\Gamma|^2 + |\Gamma| e^{j(2\beta z + \theta_\Gamma)} + |\Gamma| e^{-j(2\beta z + \theta_\Gamma)}}$$

$$|V(z)| = |V_o^+|\sqrt{1 + |\Gamma|^2 + 2|\Gamma| \cos(2\beta z + \theta_\Gamma)}$$

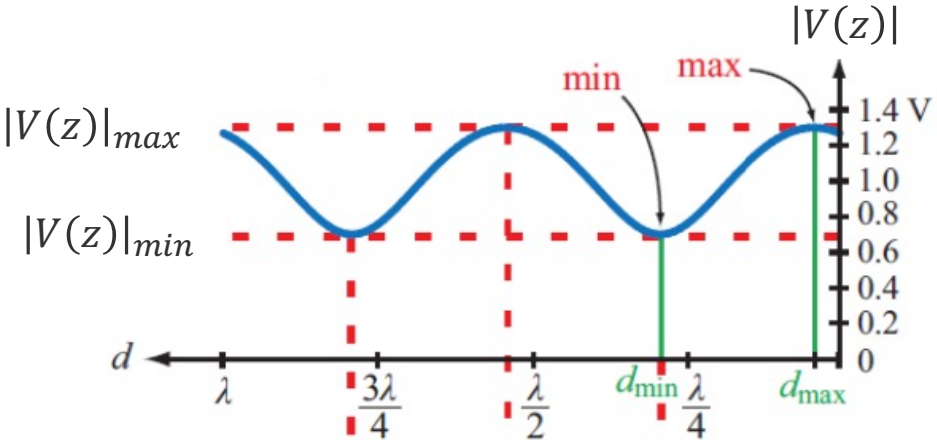
$$|V(z)|_{max} = |V_o^+|\,(1 + |\Gamma|)$$

$$|V(z)|_{min} = |V_o^+|\,(1 - |\Gamma|)$$

$$S = \frac{|V(z)|_{max}}{|V(z)|_{min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

Voltage Standing Wave Ratio (VSWR)

A measure of how good is the load matching.
VSWR=1 perfect match

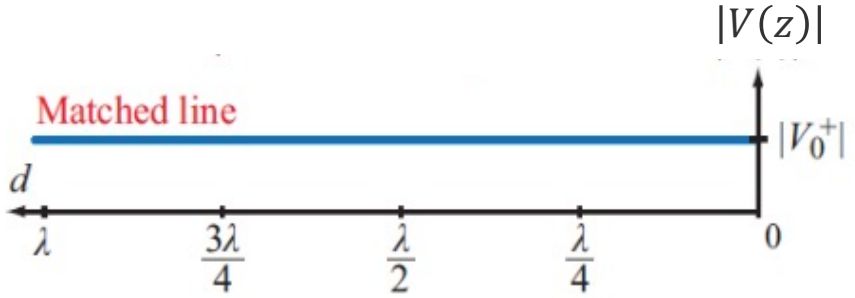


Special Cases

1) Matched Line

$Z_L = Z_o$

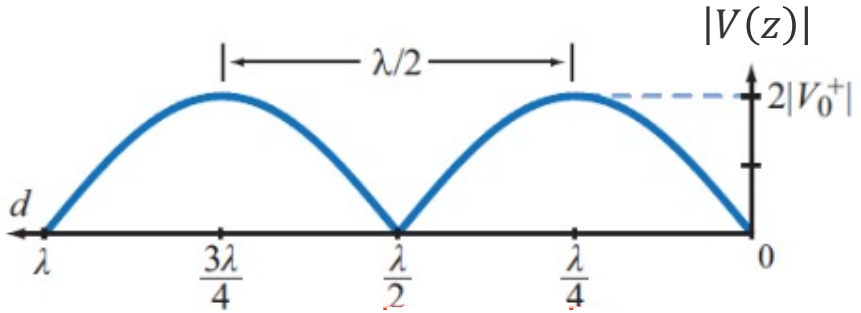
$\Gamma = 0$



2) Shorted Line

$Z_L = 0$

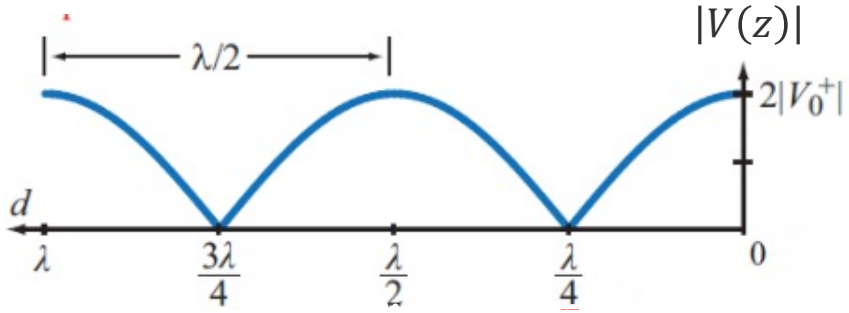
$\Gamma = -1$



3) Open-circuited Line

$Z_L = \infty$

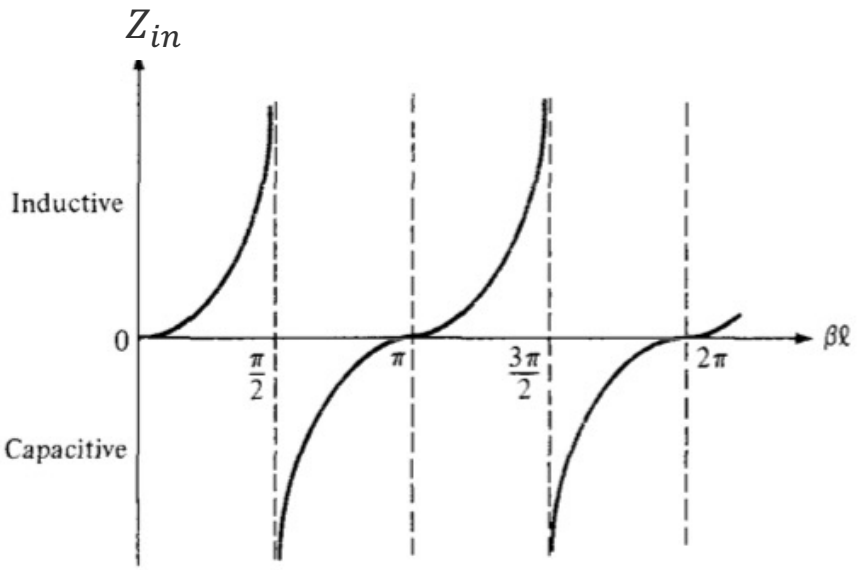
$\Gamma = 1$



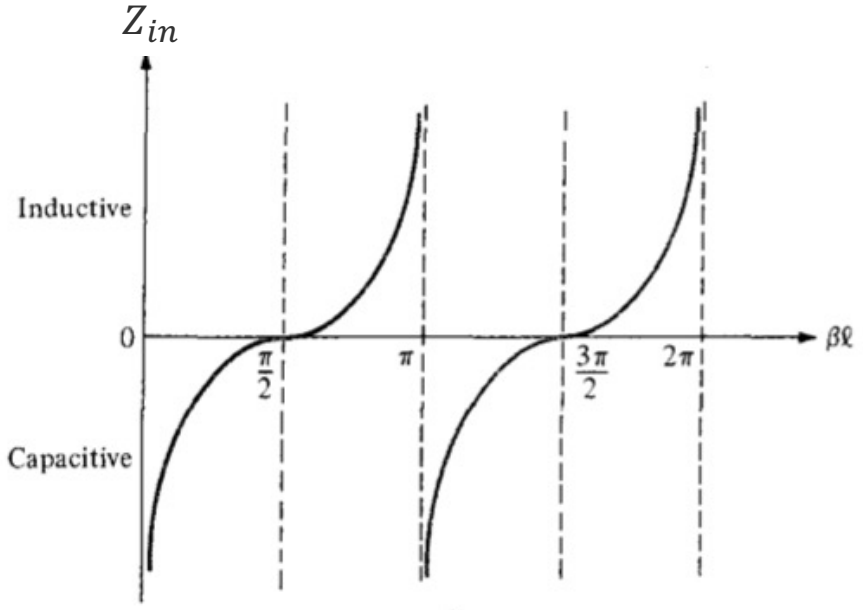
Special Cases

Input impedance versus line length for the two special cases.

$$Z_{in} = Z_o \left[\frac{Z_L + jZ_o \tan(\beta l)}{Z_o + jZ_L \tan(\beta l)}\right]$$



Shorted Line



Open-circuited Line

Power Analysis

Starting from the equations, $V(z) = V_o^+ \left(e^{-j\beta z} + \Gamma\, e^{j\beta z}\right)$ $\quad I(z) = \dfrac{V_o^+}{Z_o}\left(e^{-j\beta z} - \Gamma\, e^{j\beta z}\right)$

In general, $V_o^+ = |V_o^+| e^{j\phi}$

We need instantaneous expressions.

$$v(z,t) = |V_o^+|\left[\cos(\omega t + \beta z + \phi) + |\Gamma|\cos(\omega t - \beta z + \phi + \theta_\Gamma)\right]$$

$$i(z,t) = \frac{|V_o^+|}{Z_o}\left[\cos(\omega t + \beta z + \phi) - |\Gamma|\cos(\omega t - \beta z + \phi + \theta_\Gamma)\right]$$

$$P(z,t) = \frac{|V_o^+|^2}{Z_o}\left[cos^2(\omega t + \beta z + \phi) - |\Gamma|^2 cos^2(\omega t + \beta z + \phi + \theta_\Gamma)\right]$$

Incident power Reflected power

$$P^i(z,t) = \frac{|V_o^+|^2}{2Z_o}\left[1 + \cos(2\omega t + 2\beta z + 2\phi)\right]$$

DC AC

$$P^r(z,t) = -|\Gamma|^2 \frac{|V_o^+|^2}{2Z_o}\left[1 + \cos(2\omega t + 2\beta z + 2\phi + 2\theta_\Gamma)\right]$$

DC AC

$$P^i{}_{ave} = \frac{|V_o^+|^2}{2Z_o}$$

$$P^r{}_{ave} = -|\Gamma|^2 \frac{|V_o^+|^2}{2Z_o} = -|\Gamma|^2\, P^i{}_{ave}$$

$$P_{ave} = \frac{|V_o^+|^2}{2Z_o}(1 - |\Gamma|^2)$$

Independent of location along the line

Examples

(1) A 30-m-long lossless transmission line with $Z_o = 50$ Ohms at 2 MHz is terminated by a load with $Z_L = 60 + j40$ Ohms. The wave velocity on the line is $0.6c$.
Find Γ_L, $VSWR$ and Z_{in}.

$$\Gamma_L = \frac{Z_L - Z_o}{Z_L + Z_o} = \frac{60 + j40 - 50}{60 + j40 + 50} = 0.3523 \angle 56°$$

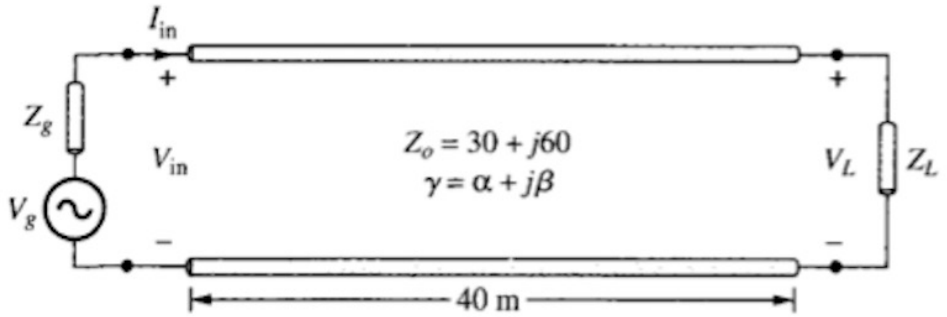
$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1 + 0.3523}{1 - 0.3523} = 2.088$$

$$\beta = \frac{\omega}{v} \quad \Rightarrow \quad \beta l = \frac{\omega l}{v} = \frac{2\times10^6\times30}{0.6\times3\times10^8} = \frac{2\pi}{3}$$

$$Z_{in} = Z_o\left[\frac{Z_L + jZ_o\tan(\beta l)}{Z_o + jZ_L\tan(\beta l)}\right] = 50\frac{(60 + j40) + j50\tan 120}{50 + j(60 + j40)\tan 120} = 24 \angle 3.22° \ \Omega$$

Examples

(2) A 40-m-long transmission line with $Z_o = 30 + \text{j}60$ Ohms. The load is matched to the line. Given $Z_g = 30 + \text{j}60$ Ohms, $V_g = 15\angle 0\, V(\text{rms})$ and $V_L = 5\angle -48°\, V(\text{rms})$, find Z_{in}, I_{in}, V_{in} and γ.



$$Z_L = Z_o$$

$$Z_{in} = Z_o\left[\frac{Z_L + jZ_o\tan(\beta l)}{Z_o + jZ_L\tan(\beta l)}\right] = Z_o = 30 + \text{j}60\ \Omega$$

$$I_{in} = \frac{V_g}{Z_{in} + Z_g} = \frac{15\angle 0}{2(30 + \text{j}60)} = 0.112\angle -63.4°\ (\text{A})$$

$$V_{in} = \frac{V_g}{2} = 7.5\angle 0\ (\text{V rms})$$

$$V_o^+ = \frac{1}{2}(V_{in} + I_{in}Z_o) = \frac{1}{2}(V_L + Z_oI_L)\ e^{\gamma l} \quad \Rightarrow \quad e^{\gamma l} = \frac{V_{in} + I_{in}Z_o}{V_L + Z_oI_L}$$

$$\gamma = 0.0101 + 0.2094j \quad (1/\text{m})$$

Q & A

Send me an email if you have any questions.